

ARTIFICIAL INTELLIGENCE INTERNSHIP

Internship Presentation

Ayush Dixit

Artificial Intelligence Intern | Codec Technologies Pvt. Ltd.

28 May 2026 - 29 August 2026 | 3 Month AICTE & ICAC Approved Internship

Internship at a Glance

Organization

Codec Technologies Pvt. Ltd.

Role

Artificial Intelligence Intern

Duration

3 Months • 28/05/2026 - 29/08/2026

Approval

AICTE & ICAC Approved Internship Program

Program Manager

Dr. Anurag Shrivastava

Internship Objectives

Build practical understanding of Artificial Intelligence workflows.

Connect academic concepts with industry-oriented problem solving.

Strengthen skills in data preparation, model development and evaluation.

Practice structured experimentation, documentation and presentation.

Develop professional habits: version control, reproducibility and communication.

Artificial Intelligence Workflow

1. Problem Definition

Understand the business or technical objective.

2. Data

Collect, inspect, clean and prepare relevant data.

3. Modeling

Select an appropriate ML/AI approach and train model.

4. Evaluation

Measure performance using suitable metrics.

5. Iteration

Analyze errors and improve the solution.

6. Delivery

Document results and communicate the outcome.

Data Preparation & Exploration

Understand dataset structure, variables, data types and missing values.

Clean inconsistent or duplicate records before modeling.

Explore distributions and relationships to identify useful patterns.

Transform features into a model-ready representation.

Keep preprocessing steps consistent between training and evaluation data.

Machine Learning Model Development

Define the target variable and relevant input features.

Split data into training and testing sets to estimate generalization.

Choose baseline and candidate algorithms according to the problem.

Train models and tune important parameters where appropriate.

Compare alternatives using consistent evaluation procedures.

Model Evaluation

Accuracy

Overall proportion of correct predictions.

Precision

How many predicted positives are actually positive.

Recall

How many actual positives are successfully identified.

F1 Score

Balance between precision and recall.

Confusion Matrix

Detailed view of correct and incorrect classifications.

Error Analysis

Study failure cases to guide improvement.

Tools & Technologies

Python

Core programming language for AI/ML exper

NumPy / Pandas

Numerical computing and structured data and

Matplotlib

Visualization for exploratory analysis and results.

Scikit-learn

Common machine-learning algorithms and ev

Git / GitHub

Version control, collaboration and project sharing.

Practical Learning & Problem Solving

Translated theoretical AI concepts into a structured practical workflow.

Focused on understanding why each preprocessing and modeling step is required.

Used evaluation results to reason about strengths, limitations and possible improvements.

Learned to present technical work clearly through organized documentation.

Developed a repeatable approach: prepare → train → evaluate → improve → document.

Challenges & Solutions

Data Quality

Challenge: incomplete or inconsistent data.

Solution: systematic inspection and preprocessing.

Model Selection

Challenge: choosing an appropriate baseline.

Solution: compare candidate approaches using the same metrics.

Overfitting

Challenge: strong training results but weaker unseen-data performance.

Solution: validation, regularization and careful feature/model choices.

Communication

Challenge: explaining technical results simply.

Solution: structured visuals, metrics and concise conclusions.

Key Outcomes & Skills Developed

Artificial Intelligence and machine-learning workflow awareness.

Python-based data analysis and experimentation skills.

Understanding of model evaluation and error analysis.

Technical documentation and presentation skills.

Version-control and GitHub repository practices.

Improved analytical thinking and structured problem solving.

Professional Takeaways

AI development is iterative: good results come from repeated measurement and improvement.

Data quality and preprocessing strongly influence model reliability.

Evaluation should reflect the actual problem rather than relying on one metric.

Clear documentation makes technical work easier to review, reproduce and extend.

Industry learning complements academic knowledge by emphasizing practical execution.

Internship Certification

Certificate awarded to Ayush Dixit for completing the internship program.

Organization: Codec Technologies Pvt. Ltd.

Role: Artificial Intelligence Intern.

Program: 3 Month AICTE & ICAC Approved internship.

Conducted from 28/05/2026 to 29/08/2026.

Program Manager: Dr. Anurag Shrivastava.

THANK YOU

Ayush Dixit

Artificial Intelligence Intern

Codec Technologies Pvt. Ltd.